Gaurav Chavda

MERN Stack Developer

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PROFESSIONAL SUMMARY

MERN Stack Developer with hands-on experience building scalable, responsive, and secure web applications using React.js, Node.js, Express.js, and MongoDB. Skilled in RESTful API development, authentication and authorization, reusable component architecture, and real-time features. Focused on clean, maintainable code and delivering reliable, user-friendly solutions. Experienced in integrating frontend and backend systems, database-driven functionality, and secure application workflows. Strong problem-solving and debugging skills with a focus on performance, code quality, and continuous learning.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, React Hooks, React Router, Redux Toolkit, Material UI (MUI), Bootstrap, Vite, Axios, Formik, Yup

Backend & APIs: Node.js, Express.js, RESTful APIs, JWT Authentication, Role-Based Access Control (RBAC), Swagger/OpenAPI, Multer, Nodemailer

Real-Time: Socket.IO, WebRTC

Database: MongoDB, MySQL

Tools & technologies: Git, GitHub, Postman, VS Code, npm

INTERNSHIP EXPERIENCE

Virtual Height | Ahmedabad, India

MERN Stack Developer Intern (6-Month Internship) March 16, 2026 – September 30, 2026

- **Full-Stack Development:** Contributed to Solar Management System and Real-Time Chat Application, working across frontend, backend, database, and API integration.
- **Frontend Development:** Built 40+ reusable React.js components including forms, data tables, modals, filters, and responsive interfaces across multiple application modules.
- **Backend Development:** Developed and integrated RESTful APIs for CRUD operations, business workflows, authentication, authorization, and database-driven features.
- **Development & Collaboration:** Tested and debugged application features, resolved frontend/backend integration issues, and followed Git/GitHub-based team development practices.

PROJECTS

Bidirectional — Real-Time Chat & Video Calling Platform Demo

Technologies: *React.js, TypeScript, Node.js, Express.js, Socket.IO, Redis, WebRTC, MongoDB, JWT, Axios, Vite, Swagger*

- Built a full-stack real-time chat application using React.js (custom Hooks), TypeScript, Node.js, Express.js, Socket.IO (WebSockets) and MongoDB, with 1:1 and group chat, typing indicators, read receipts, reactions, replies, forwarding, editing, deletion and GIF sharing.
- Implemented real-time voice/video calling with Socket.IO signaling and WebRTC, including screen sharing, device switching, busy-user detection and ICE candidate handling.
- Developed JWT authentication with refresh-token rotation, bcrypt password hashing, rate-limited authentication endpoints and OTP-based email verification (Nodemailer).
- Built a consent-based screen-sharing workflow with user approval and live WebRTC status updates.
- Designed MongoDB indexes and RESTful APIs with Swagger/OpenAPI documentation, along with image processing, upload validation and privacy-aware presence tracking.
- **Scalable Architecture & Redis Adapter:** Integrated Redis Pub/Sub adapter with Socket.IO to enable multi-instance horizontal scaling and distributed event broadcasting, paired with cursor-based infinite scrolling for smooth loading of large chat histories.

Solar Installation Management System — Full-Stack ERP Application Demo

Technologies: *React.js (React 19), Node.js, Express.js, MongoDB, Mongoose, TanStack React Query, JWT, Bootstrap, Material UI, Recharts, Leaflet, React-Leaflet, Formik, Yup, Joi, Multer, Cloudinary, Nodemailer, jsPDF, XLSX, PDFKit, Helmet*

- **End-to-End Solar ERP Lifecycle:** Architected a full-scale ERP using React 19, Node.js, Express, and MongoDB to digitize the entire solar installation lifecycle—from lead management, rooftop site surveys, and dynamic quotation generation to project approvals, commissioning, subsidy tracking, and AMC service management.
- **Server-State Management & Performance:** Integrated TanStack React Query for server-state caching, background refetching, and query invalidation alongside React.lazy / Suspense code-splitting to minimize redundant API calls and reduce UI latency.
- **Granular RBAC & Secure Authentication:** Implemented stateless JWT authentication stored in secure HttpOnly cookies (with refresh tokens) and custom Express rbacGuard middleware enforcing strict role-based access control across 40+ protected API modules.
- **GIS Mapping & Site Coordinates:** Embedded Leaflet / React-Leaflet with GPS geolocation and reverse geocoding to allow technicians and surveyors to map precise rooftop coordinates and installation sites interactively.
- **Optimized MongoDB & High Data Scale:** Engineered structured Mongoose data models with single and compound indexing, aggregation pipelines ($match, $group), dynamic pagination, and search filters tested against datasets of 10,000+ inventory and product records.
- **Interactive Analytics & Dual-Engine Reporting:** Built real-time KPI dashboards with Recharts and implemented dual-layer export reporting—client-side PDF (jsPDF) and Excel (XLSX) generation with on-demand chunk loading, plus server-side PDFKit cursor streaming.
- **Layered Security & Cloud Media Pipeline:** Hardened the platform with Express Rate Limiting, Helmet HTTP security headers, CORS, bcrypt, and dual-layer schema validation (Formik/Yup on frontend, Joi on backend), integrated with Cloudinary & Multer for automated media uploads and Nodemailer for automated customer email alerts.

EDUCATION

Bachelor of Computer Applications (BCA) Graduated: 2024
Shree Adarsh BCA College, Botad, Gujarat

Higher Secondary (12th) Graduated: 2022
Akshar Purushottam High School, Gujarat

Secondary (10th) Graduated: 2020
Akshar Purushottam High School, Gujarat

CERTIFICATIONS

- MERN Stack Development Training, Virtual Height (August 2025 – March 2026)
- MERN Stack Developer Internship, Virtual Height (March 2026 – September 2026)
- React JS Course, Scaler Topics (April 2026)
- Node.js Certification Course – Master the Fundamentals, Scaler Topics (June 2026)